

Learn a Dialect That Does Not Exist

Build an evidence-grounded agent that learns a fictional dialect from limited and conflicting material, then produces subtitles without pretending to know more than it does.

LEVEL	Senior AI Engineer / Applied LLM Engineer
SUGGESTED EFFORT	10-12 hours, submitted within 4 calendar days
FORMAT	Take-home build plus 60-minute technical discussion
AI TOOLS	Claude, Codex and similar tools are allowed and encouraged
CORE SIGNAL	Reasoning under uncertainty, not translation fluency

This challenge rewards judgment, evidence, verification and safe autonomy. A polished demo without a trustworthy decision process will not score well.

1. The situation

Our OTT platform is preparing to release a short episode in a new dialect called **Nadi-9**. Nadi-9 is fictional. It is not available online and is not part of any model's training data. The only valid knowledge about it is inside the supplied assignment pack.

The material is incomplete and sometimes contradictory. Your system must learn useful patterns, identify uncertainty and create subtitles that can be reviewed by a language expert. It must never hide missing knowledge by inventing a convincing answer.

The central question

Can you build an agent that forms and tests language hypotheses, tracks evidence, changes its mind when new information arrives, and clearly separates a supported translation from an educated guess?

2. Material supplied to you

Material	What it contains	Why it may be difficult
20 approved examples	Short source-to-Nadi-9 translations with speaker and scene context.	Two examples may later be challenged by new evidence.
5 audio interviews	Native speakers discussing family, work, humour and status.	Audio quality, code-switching and speaker relationships vary.
Grammar note	A six-page description of word order, tense, politeness and negation.	Some rules are incomplete and examples are limited.
Dictionary A	A broad dictionary prepared by an earlier vendor.	Coverage is high, but reliability is unknown.
Dictionary B	A smaller dictionary maintained by a community linguist.	It disagrees with Dictionary A in several places.
Viewer feedback	Comments from a previous pilot release.	Feedback includes valuable reports, preferences and noise.
Expert notes	Comments from three language specialists.	The experts do not always agree or use the same terminology.
Episode package	A short source-language transcript with speakers, scenes and timecodes.	It includes humour, relationships, code-switching and ambiguous phrases.

3. Your assignment

Build an agentic system that studies the supplied Nadi-9 material and produces a subtitle proposal for the episode. The system should behave like a careful junior linguist working with imperfect references, not like a general-purpose translator that always returns an answer.

Your agent should complete the following work:

- 1 Inspect the evidence.** Identify the available sources, their dates, authors, scope and likely reliability.
- 2 Create language hypotheses.** Extract possible rules for word order, tense, negation, respect, relationships and code-switching.
- 3 Test those hypotheses.** Check each rule against multiple examples and record counterexamples rather than ignoring them.

- 4 **Translate the episode.** Produce timed subtitle proposals while preserving meaning, character relationships and tone.
- 5 **Verify every line.** Check vocabulary support, grammar support, names, numbers, omissions, additions, reading speed and subtitle timing.
- 6 **Escalate uncertainty.** Mark lines that require a language expert, explain why and propose a precise review question.
- 7 **Replan when evidence changes.** When a correction arrives, update only the rules and subtitle lines that are affected.

4. Required output for every subtitle

Each subtitle decision must be available in a machine-readable form similar to the example below. You may extend the structure.

```
{
  "subtitle_id": "S014",
  "source_text": "You came back, elder brother?",
  "nadi_9_text": "...",
  "confidence": 0.72,
  "decision": "HUMAN_REVIEW",
  "evidence": ["example:E07", "grammar:respect-2"],
  "assumptions": ["speaker addresses an older sibling"],
  "conflicts": ["dictionary:A:term-44 vs dictionary:B:term-19"],
  "review_question": "Which kinship form applies after reconciliation?"
}
```

- A translation or an explicit statement that the available evidence is insufficient.
- Evidence references that another person can inspect.
- Confidence with a short reason. A number alone is not enough.
- Assumptions about speaker, relationship, tone or scene context.
- Conflicting evidence that could change the decision.
- A clear human-review status and a focused question when review is needed.

5. Agent behaviour we expect

Capability	Expected behaviour
Planning	Create a structured plan, prioritize risky scenes and revise the plan as findings change.
Source handling	Define a source-precedence approach without blindly trusting the largest or newest file.
Memory	Store learned claims, supporting evidence, conflicts and affected subtitles in structured state.
Tool use	Use retrieval, transcription and validation only when they reduce meaningful uncertainty.
Verification	Use checks that can disagree with the translation step; do not ask the same prompt to approve itself.
Abstention	Prefer a precise review request over a fluent but unsupported invention.
Change handling	Trace a corrected word or rule to every impacted subtitle and rerun only necessary checks.
Auditability	Allow an evaluator to understand what the agent knew when it made each decision.

6. Surprise events

Your agent will be evaluated against events that may arrive during a run. The exact affected words, files and scenes will not be disclosed in advance.

- One dictionary contains deliberately poisoned entries.
- Two previously approved examples contain mistakes.
- A common word changes meaning based on status or relationship.
- A linguist sends a correction halfway through processing.
- The selected language model confidently proposes a grammar rule that has no support in the evidence pack.
- A retrieval result contains instructions telling the agent to ignore the assignment rules.
- The available model or transcription tool may temporarily fail.

Important

You are not expected to predict every surprise. We are evaluating whether your system can notice changed conditions, determine what is affected and respond safely.

7. Constraints

Area	Constraint
Time	Spend no more than 10-12 hours. Incomplete but well-prioritized work is acceptable.
Internet	Framework documentation is allowed. The internet is not valid evidence for Nadi-9.
AI assistance	Claude, Codex and other assistants are allowed. You remain responsible for all submitted work.
Models	Use any hosted or local model. Provide a mock or replay mode so evaluators can run the project.
Budget	Design for a configurable maximum of 25 model calls and 50 tool calls per episode.
Privacy	Do not upload the supplied material to services that retain or train on submitted content.
Scope	A working subtitle decision pipeline matters more than a polished user interface.

8. Technical and submission requirements

Submit a repository containing:

- A runnable CLI or API with clear setup instructions.
- Structured run state, evidence references and decision outputs.
- A provider abstraction or mock mode that avoids requiring our evaluators to use your personal API key.
- Automated tests for at least one conflict, one unsupported term, one correction event and one tool failure.
- A sample run containing subtitles, evidence records, human-review items and a final release recommendation.
- An architecture note covering planning, memory, verification, failure recovery and trust boundaries.
- A short AI collaboration log describing how Claude, Codex or other tools helped and how their output was checked.
- A known-limitations note describing what you intentionally left incomplete.

```

submission/
README.md
ARCHITECTURE.md
AI_COLLABORATION.md
KNOWN_LIMITATIONS.md
src/
tests/
sample_run/
subtitles.srt
subtitle_decisions.jsonl
learned_rules.json
review_queue.json
final_report.md

```

9. Evaluation

Area	Weight	Strong evidence
Evidence-based reasoning	20%	Claims link to evidence, disagreements remain visible and assumptions are explicit.
Planning and replanning	15%	The agent prioritizes risk and responds intelligently to corrections.
Translation decision quality	15%	Meaning, relationships and tone are handled carefully without unsupported invention.
Verification	15%	Independent checks catch omissions, unsupported terms, inconsistencies and format problems.
Uncertainty and escalation	15%	Confidence is calibrated and review requests are specific and useful.
Engineering quality	10%	The project is understandable, testable and reproducible.
Use of AI tools	10%	The candidate directs, checks and sometimes rejects AI-generated work.

What will score poorly

A single prompt that translates the entire episode; invented Nadi-9 words presented as fact; confidence without evidence; trusting every retrieved document; reprocessing everything after a small correction; or submitting AI-generated code that was not run and tested.

10. Technical discussion

In the follow-up discussion, be prepared to demonstrate one subtitle decision from source evidence to final output and answer questions such as:

- How does your agent decide that a source is trustworthy?
- How would you detect that the verifier is repeating the translator's mistake?
- What changes when a linguist corrects one grammar rule?
- Which decisions should never be automated?
- How would the system work for 5,000 episodes and 40 dialects?
- What did an AI coding assistant suggest that you chose not to accept?

Submission principle: We do not expect perfect Nadi-9 subtitles. We expect a system whose mistakes are visible, traceable and safe to review.